SentimentEdge: Multi-Agent Architecture

Phase 2 · Track A · Five-Agent Sequential Pipeline

Sequential pipeline: data loads once, each agent waits for its predecessor, the Filter acts as a hard gate, and the pipeline stops on zero results. State is per-run only (no cross-session persistence). All outputs are saved to a timestamped run directory.

LEGEND Data collection · Hard gate / filtering · LLM inference (Claude API) · Aggregation / routing · Output / reporting · Stopping condition · Human-in-the-loop

INPUT DATA (JSONL)
posts.jsonl *(6,919 raw posts · r/wallstreetbets)*
comments.jsonl *(945,096 comments)*

↓

1 COLLECTOR

Collector Agent

Loads posts and comments from the JSONL dataset files into structured pandas DataFrames. Validates file paths and confirms row counts. No filtering, transformation, or API calls at this stage; raw data only.

READS
posts.jsonl (6,919 rows)
comments.jsonl (945,096 rows)

WRITES
df_posts_raw
df_comments_raw

DATA INGESTION · NO API CALLS · NO FILTERING

df_posts_raw + df_comments_raw passed to Filter Agent

↓

2 FILTER AGENT

HARD GATE

Filter Agent

Hard gate before any LLM calls. Merges bundled top comments onto their parent posts, applies three filter rules in sequence, computes engagement scores, and builds the llm_text field that Claude receives. No post reaches the Sentiment Agent unless it passes every rule.

FILTER RULES (APPLIED IN ORDER)
- x Remove posts with no body AND no bundled comments (handles the 64% deleted-post problem
- x Remove posts with upvote score below 10
- x Remove stickied moderator posts

READS
df_posts_raw · df_comments_raw
min_score threshold (config)

WRITES
df_filtered: 1,458 posts (21.1% of raw)
+ engagement_score · llm_text columns

ENGAGEMENT = SCORE X 0.7 + NUM_COMMENTS X 0.3

STOPPING CONDITION
If df_filtered = 0 posts after all filter rules, the pipeline halts immediately. Metadata and partial state are written to trace.txt and run_metadata.json. No API calls are made.

df_filtered (1,458 posts; 500 sampled) passed to Sentiment Agent

↓

3 SENTIMENT · CLAUDE API

Sentiment Agent

Sends each post's llm_text to claude-sonnet-4-20250514 via the Anthropic Messages API. The static system prompt is cached with cache_control: ephemeral across the batch, reducing input token costs by roughly 90%. Each post returns a 13-field JSON object. Parse failures return neutral defaults (confidence 0.0) and are logged rather than silently skipped.

sentiment · confidence · tickers · primary_ticker · emotions · post_type · is_sarcastic · market_relevance · key_insight · is_rumour · rumour_type · rumour_confidence · rumour_summary

READS
df_filtered (per-post llm_text)

WRITES
df_llm: one row per post

STOPPING CONDITION
If the Sentiment Agent returns 0 valid rows (all parse failures), the pipeline halts. JSON parse failures return neutral defaults (confidence 0.0) and are logged. Partial state is preserved in trace.txt.

Anthropic API key (env var)
batch_size (config, default 50-100)

13 fields per row from Claude
parse errors logged to trace.txt

PROMPT CACHING ON · 500 POSTS ANALYZED · PARSE FAILURES DEFAULT TO NEUTRAL (CONF 0.0)

df_llm (per-post 13-field results) passed to Aggregator

4 AGGREGATOR · TWO-TRACK ROUTER

Aggregator Agent

Groups per-post LLM results by primary_ticker (engagement-weighted). Routes posts with high-confidence rumour flags to a separate output track. Adds a monthly trend column to df_llm for the Output Agent. Saves all three CSVs to the timestamped run directory. Posts with primary_ticker = UNKNOWN are excluded from the ticker summary but remain in df_llm.

READS
df_llm (all analyzed posts)
rumour_threshold (config, 0.70)
low_conf / low_sample thresholds

WRITES
Track 1: ticker_summary.csv (68 tickers)
Track 2: rumour_alerts.csv (conf >= 0.70) · 7 flagged
df_llm + month column added

TRACK 1: TICKER SUMMARIES · TRACK 2: RUMOUR ALERTS

TRACK 1: TICKER SUMMARIES
ticker_summary.csv · 68 tickers found
bullish / bearish / neutral % per ticker
avg_confidence · avg_relevance · sarcastic_count
avg_engagement · low_confidence flag · low_sample flag

TRACK 2: RUMOUR ALERTS
rumour_alerts.csv · 7 posts flagged (conf >= 0.70)
Filter: is_rumour = True AND rumour_confidence >= 0.70
Labelled as unverified social media speculation.
Requires human review before any downstream use.

ticker_summary + rumour_alerts + df_llm (with month) passed to Output Agent

5 OUTPUT AGENT

Output Agent

Formats and renders the full market intelligence report to the terminal. Queries the top 3 tickers by post count, surfaces daily sentiment shift alerts, and produces a side-by-side comparison of the two leading tickers. A --replay flag re-runs the Output Agent against any saved run directory without repeating API calls.

- query_ticker(ticker): full per-ticker report with sentiment breakdown, sarcasm warning, rumour flag, and top key insights with Reddit permalinks
- run_trend_alerts(): detects daily sentiment shifts above the configured threshold and surfaces them as time-bound alerts
- compare_tickers(A, B): side-by-side bullish / bearish / neutral comparison of the two highest-volume tickers

READS
ticker_summary (from Aggregator)
rumour_alerts (from Aggregator)
df_llm with month column

WRITES
Terminal report (stdout via TeeLogger)
sentiment_results.csv
trace.txt · run_metadata.json

TERMINAL OUTPUT · REPLAY MODE VIA --REPLAY · NO ADDITIONAL API CALLS

RUN ARTIFACTS (PER TIMESTAMPED DIRECTORY)
→sentiment_results.csv: all 500 analyzed posts
→ticker_summary.csv: 68-ticker aggregate
→rumour_alerts.csv: flagged posts
→trace.txt: full terminal output log
→run_metadata.json: config snapshot and stats

ERROR HANDLING AT AGENT BOUNDARIES
→API failure: row skipped, neutral defaults returned
→JSON parse error: caught, logged, confidence set to 0.0
→Zero results: pipeline stops, metadata saved
→Partial state: preserved in trace.txt on any error

HUMAN-IN-THE-LOOP: ANALYST REVIEW

Reviews ticker sentiment reports and interprets sarcasm-rate warnings · manually validates flagged rumour alerts before any downstream action · sarcasm detection reviewed against source posts (85% confirmed in prototype run)